

for handling the alarm duration and timer2 for blinking the segments while the clock is in 'Settings' mode. After this, declare and initialise certain variables for ADC calculations, time seeking, etc. Then initialise I2C pins, read EEPROM contents for alarm values and initiate an infinite super-loop where the RTC chip registers are read and displays are updated continu-

ously. Also in the same loop, the user inputs are acknowledged and associated events like setting alarm on/off and displaying date-time are initiated. Also, the current time is compared against the alarm variables and alarm is initiated when the time is matched.

Note that while burning the hex file solarclock.hex to the microcontroller, the EEPROM file solarclock.eep must also be programmed. At Efy Lab, PROGISP programmer was used to program solarclock.hex and solarclock.eep. Fuse bits should be D9 = high and E1 = low as shown in Fig. 7.

The clock

Charging. As mentioned earlier, the clock can be charged from solar power as well as AC mains power. Install a 10-12W, 7W solar panel on the rooftop such that it receives sufficient sunlight throughout the day. Route connections from the solar panel (SP1) carefully to the clock's charging terminals, keeping in mind the correct polarity.

Solar power is more than enough to provide continuous power to the clock forever, and the clock

can work for more than 60 hours on a single charge. But when solar power is not available on cloudy days or at night, users can charge the clock using 230V AC mains power. The solar panel rail is automatically cut off while charging through AC mains power. This way the clock never runs out of power. Switch S1 can be used to cut off all the load from the battery.

The two 3x1-watt LED modules can be individually switched on as required. Using the USB jack, you can charge all possible USB applications. Thus the project also functions as a solar-powered USB charger. The charging status is indicated by LED1 in the power unit, where a glowing LED means the battery is charging. The power unit circuit can automatically shut off the load when the battery voltage reaches the state of discharge.

Settings. The clock automatically turns on when the mains load switch (S1) is closed. The user-interface of the clock is designed such that only three buttons (S4 through S6) are required to set the various parameters like current time, hours and date-time, and to display the current date-time, set alarm and room temperature.

The Plus button can be used to increment the values in 'Settings' mode. Pressing it once during normal time display mode, first displays the room temperature and then the set

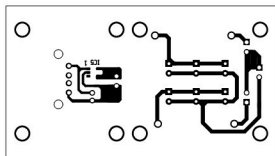


Fig. 10: An actual-size, single-side PCB of LED and USB circuit

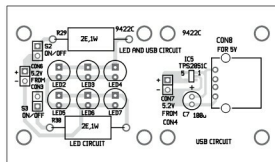


Fig. 11: Components layout for the PCB in Fig. 10

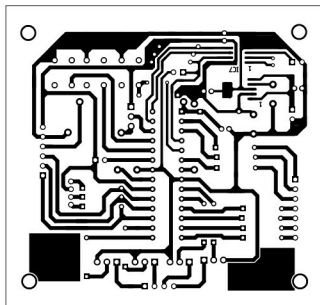


Fig. 12: Actual-size PCB layout of main clock controller circuit

